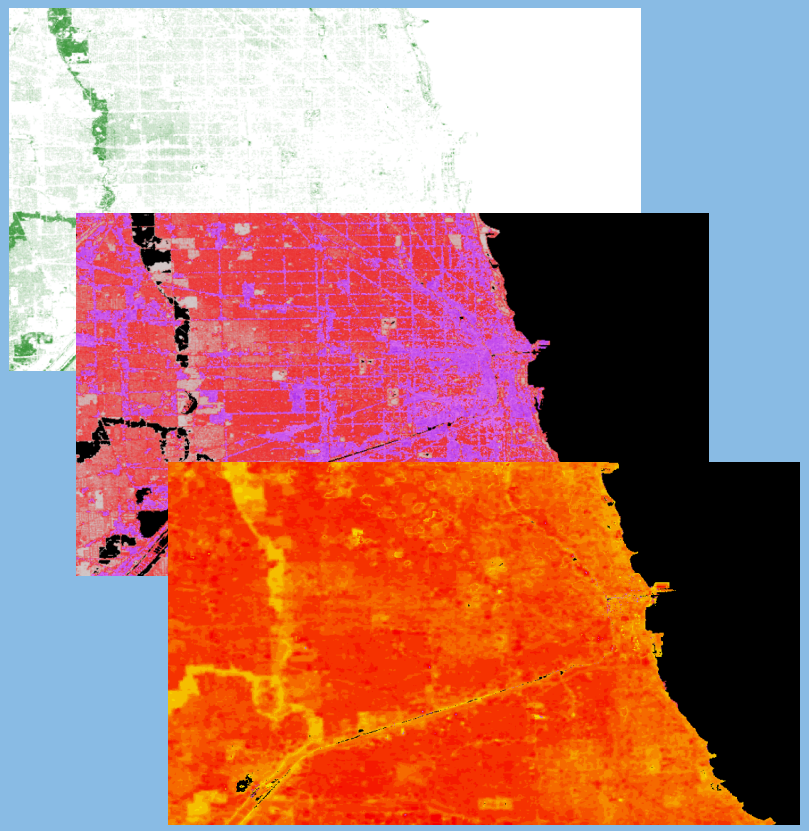
Madison Dyer

GIS Portfolio



About

I am a GIS professional with a background in environmental science. Through my bachelor’s degree in ecology and evolutionary biology at the University of Tennessee, I explored the environmental conditions that shape human experience. My particular interest is in agriculture and plant sciences.

The graduate certificate in geographic information sciences (GISc) at UNC-Chapel Hill developed my ability to investigate and model spatial data. My work focused on the interactions between infrastructure and nature. This is a showcase of several GIS projects, largely taken from my coursework. Beyond the geospatial analysis, I value engaging cartographic design and clear communication. Thank you for taking the time to explore my work.

Targeted Tree Plantings for Urban Heat Islands

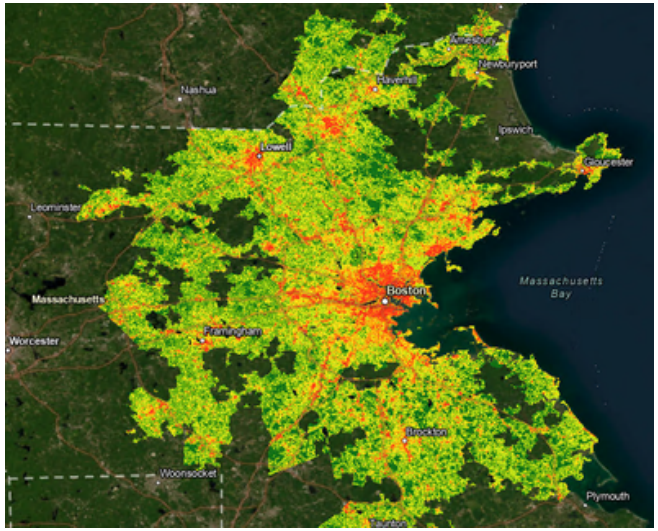
Objectives

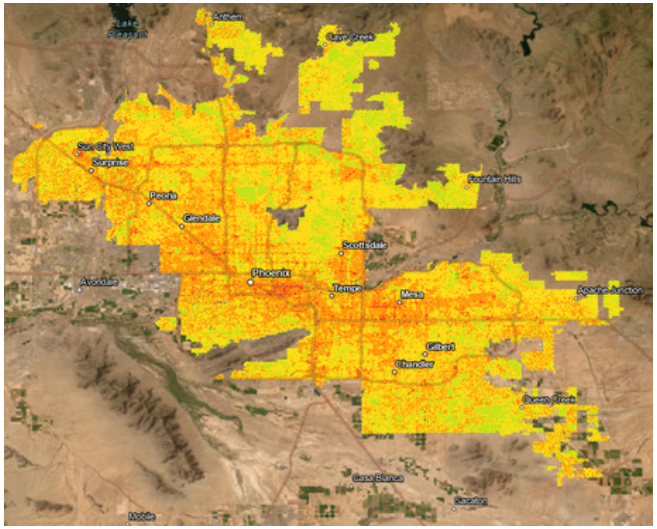
Create a Python application to be used inside ArcGIS Pro. Produce a tool that could be used by city planners to identify areas where tree cover could mitigate urban heat island effect with a Planting Priority Index (PPI).

Procedure

Our tool allows the user to select a city and produces a suitability analysis using weighted variables. For our variables, we chose to look at pre-existing canopy cover, impervious surfaces, and land surface temperature. The remote sensing data for these variables was collected using Google Earth Engine.

The framework for our program was made with Model Builder in ArcGIS Pro. We exported the model to a Python script to fine-tune the process. Each variable was reclassified to represent the need for trees, for instance, cells with high pre-existing tree canopy cover were given lower priority than cells with fewer trees. The sum of these variables gives us the final PPI. The output is a raster showing areas of high and low priority.





Planting Priority Index results: *Boston (left), shows a cluster of high PPI near the city center, while Phoenix (right) has a higher need in general, with the highest values along the I-10 corridor.*

Challenges

We wanted our tool to be applicable for cities across the country. However, given the range of climates, one scale for temperatures in our reclassification scheme would loose meaning very quickly. If we compared Phoenix and Boston with the same scale, all we would be able to learn is that Boston is cooler than Phoenix. Python scripting helped us tackle this issue. For every city, we would find the average temperature and the standard deviation. We could then create a unique reclassification scheme using deviation from the mean, where cells would be compared to the average of the whole city. While this method makes it difficult to compare results of different cities, within one city, it makes areas of anomalously high temperatures clear to the user.

Outcome

This tool is helpful for consolidating and summarizing multiple variables that contribute to urban heat islands. At a broad scale, it helps to see general trends across a city. Our example of two neighborhoods in Chicago, Hyde Park and Cicero, show that access to green space is uneven between neighborhoods.

As heat is a leading environmental cause of illness, addressing the lack of green space, therefore mitigating urban heat islands, may be a strategy for landscape managers to investigate.

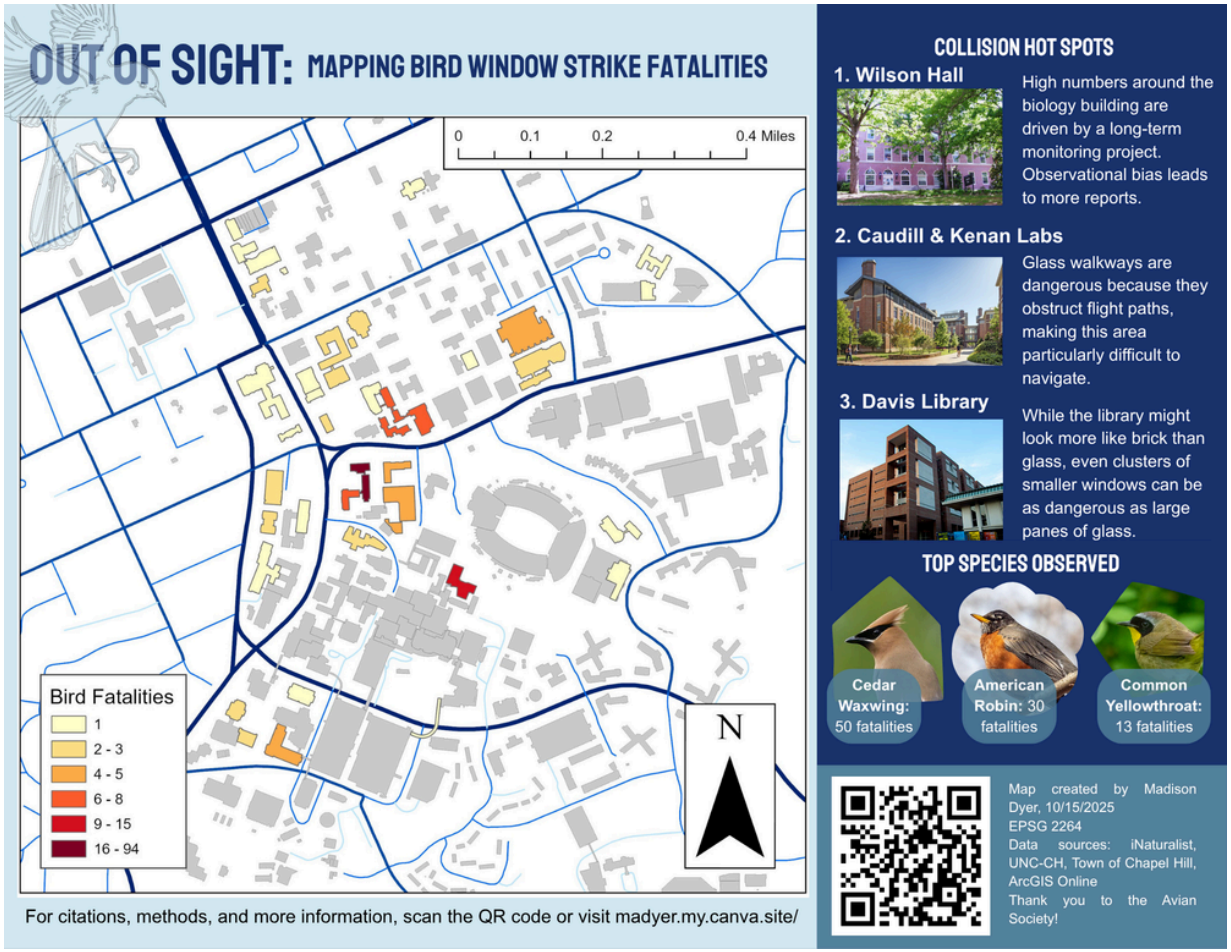


Comparing PPI in Chicago: *Green cells in Hyde Park (left) show lower PPI values than Cicero (right), where cells are mostly red or orange.*

Bird Window Strike Fatalities

Objectives

Create an informative and eye-catching graphic for UNC's GIS Day mapping competition. Communicate an issue in conservation for a popular audience with limited space.



Procedure

The UNC Avian Society has been a part of a project to map bird window-strike fatalities on campus. Sightings are reported on iNaturalist. I downloaded this data and found a map of UNC buildings. Using ArcGIS Pro, I found the nearest building for each window-strike record. Then, the bird fatalities were counted up for each building. While the map was created in ArcGIS Pro, I created the graphic in Canva. I wanted people to be able to see my sources, so I created a simple website using Canva's site feature. This also allowed me to add a little more analysis. I found the species and seasonality patterns using a pivot table in Excel.

Challenges

The competition was partially based on audience votes, so I wanted to create a map that would be simple but eye-catching. The rules of the competition meant it had to fit on a standard 8.5”x11” sheet of paper. I kept the map very simple, and I focused on UNC’s campus so the audience would be familiar with the area without me having to add context. Adding the website portion helped me use my space efficiently. I could put the most striking things on the paper, while having extra information online.



Outcome

My map won first place in the competition! Plus, I received a lot of positive feedback. People mentioned that they liked the simplicity of the map, they found it easy to understand. They also enjoyed the extra layer of detail provided by the website. The Avian Society has been advocating for preventative measures against window-strikes around campus. My hope is that bringing this issue to a wider audience with clear recommendations will help build momentum for their efforts.



Samples from the website: *Since the size of the map was restricted to an 8.5"x11" paper, I used a website to give myself more space for data visualizations.*

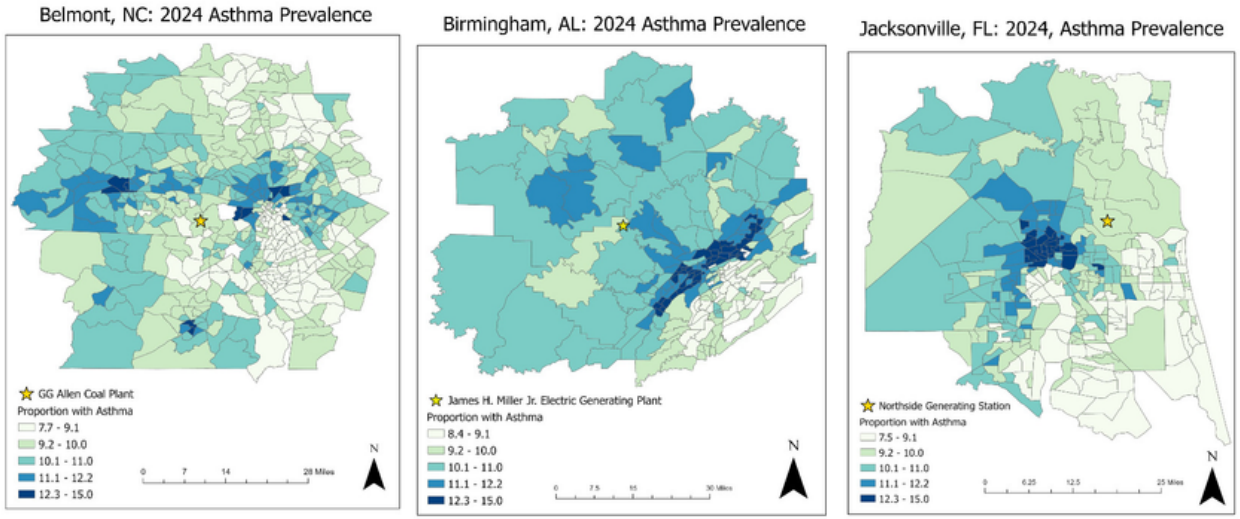
Coal Power and Respiratory Illness

Objectives

Use ArcGIS Pro to explore the spatial relationship between coal-burning power plants and respiratory health in the Southeast US using free, publicly available data.

Procedure

We used data from the CDC's PLACES survey and EIA's map of power plants. To narrow our scope, we chose power plants within 30 miles of a city. Asthma and COPD were chosen as proxies for respiratory health. For each census tract in the 30 mile buffer around the power plant, we compared the health data with the distance to the power plant and then created a linear regression model. To explore demographic influences, we also compared the health data with race and income in each tract. We found very little correlation between distance and respiratory illness, but race and income both showed higher correlation values.



Asthma prevalence for census tracts in each city: *While we did not observe a correlation with distance to the power plant, it does appear that high asthma prevalence comes in clusters.*

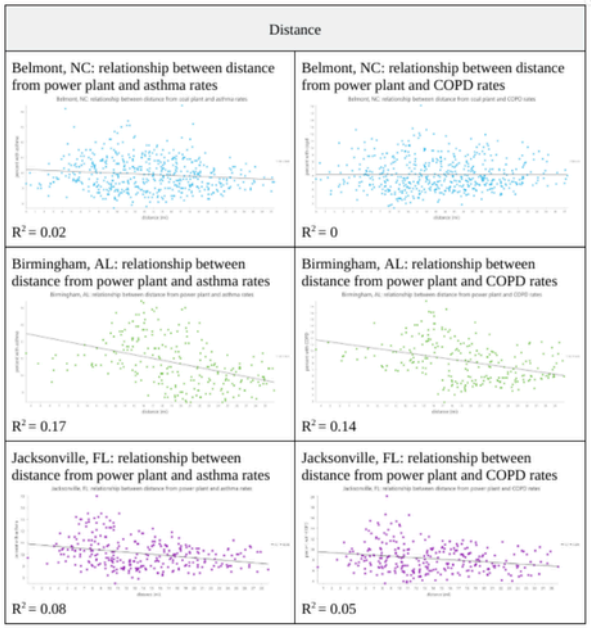
Challenges

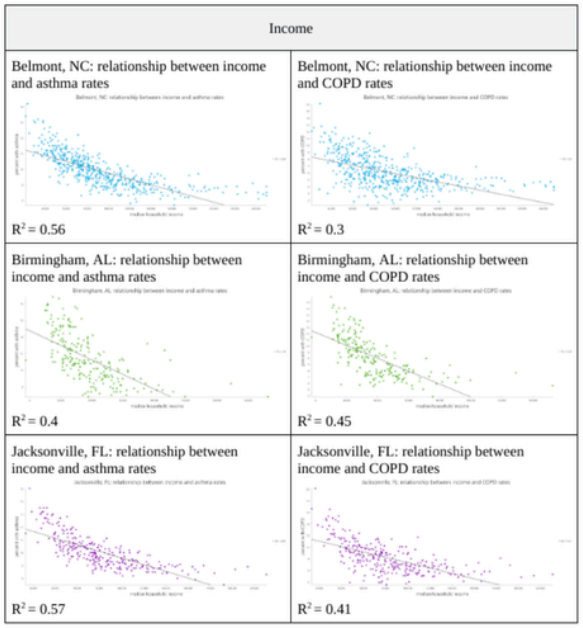
Our original goal was to investigate environmental impacts on neighborhood-level health outcomes. Looking back, cluster analysis would have helped us answer those questions. However, this class focused regression analysis, and clustering was not something we went over. We were somewhat limited by the tools available to us.

Our data was also a major limitation. We have no way of knowing whether the power plants are the main source of air pollution. Even if it was, asthma and COPD are not perfect measures of the impact of that pollution. Individual variables, like genetic predisposition or smoking, would almost certainly be better predictors of respiratory health. The data available to us was too general to tell us much, but it was enough to open up some interesting questions.

Outcome

Our project was an interesting exploration into spatial distributions of health. We did not find any evidence that individuals living closer to coal-burning power plants were any more likely to have asthma or COPD. Instead, areas with higher populations of Black individuals or lower median household incomes are more correlated with higher rates of respiratory illness. To better understand this correlation, we could investigate how areas of higher asthma and COPD prevalence are spatially arranged.





Linear regression results: *Correlation coefficient values for all regression models including distance were relatively low, between R^2 values between 0 and 0.17. Other variables had higher R^2 values, with income and asthma rates in Belmont at 0.56 as the highest correlation.*

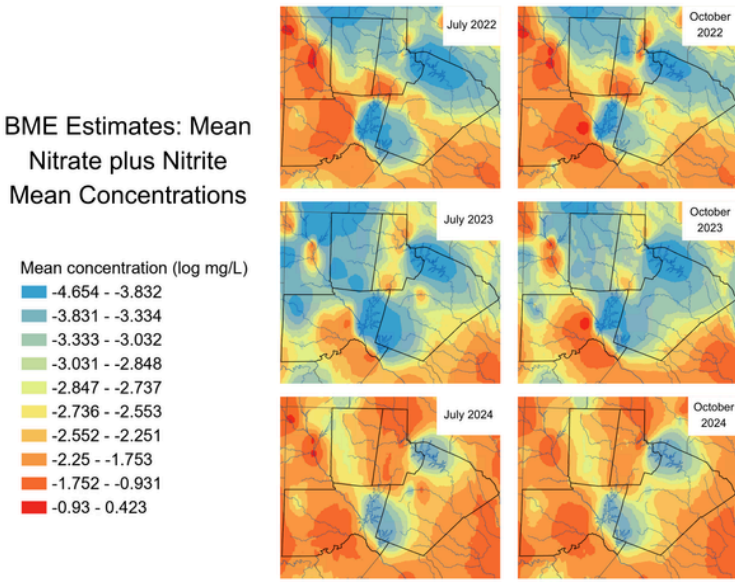
Modeling Nutrient Pollution in North Carolina’s Waterways

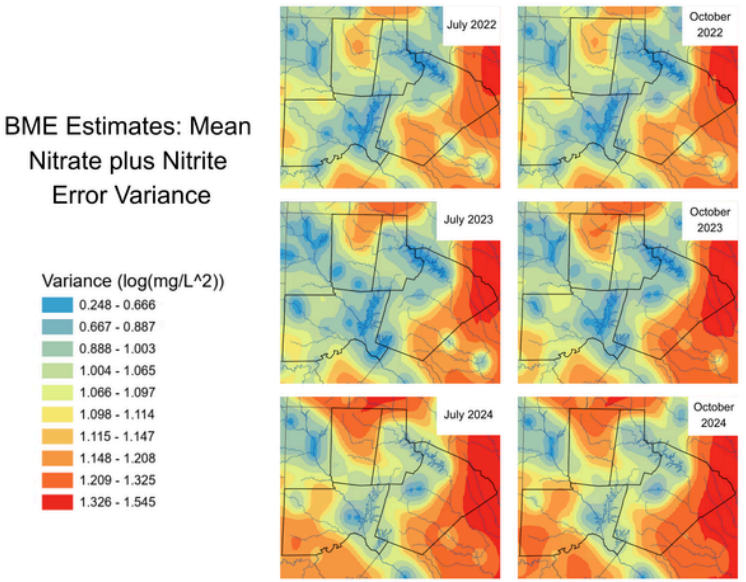
Objectives

Create a temporal and spatial model to estimate nutrient pollution levels in North Carolina using USGS monitoring data.

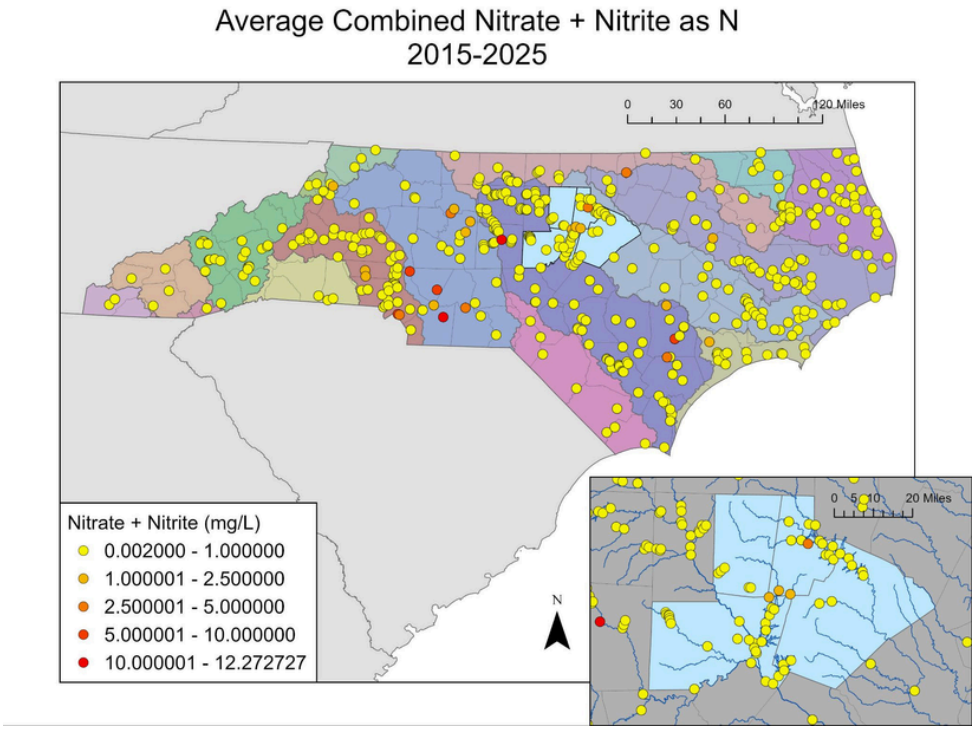
Procedure

Data was sourced from USGS surface water monitoring stations from 2020 to 2025. We chose to look at Nitrate plus Nitrite as N. Using a program developed by the professor’s lab, we created a Bayesian Maximum Entropy model with time and space as variables. The output from that model was mapped using ArcGIS Pro’s interpolate function. We found a fluctuation at a short timescale, perhaps associated with rainfall events measured in days, as well as a longer scale of fluctuations over months. Spatially, there are noticeable areas where readings remain high or low over time. Environmental conditions in those locations may be affecting the nitrogen cycle.





Model results: *Areas of high contaminants are consistently found around at the point where northern Wake County meets Durham County. Areas of blue around Falls Lake and Jordan Lake are consistently low.*



Exploration of monitoring sites across NC: *Each point shows the average reading at that location across 10 years. Concentrations greater than 0.5 mg/L are considered hazardous to human and environmental health.*

Challenges

Visualization for this project was tricky. We wanted the map to easily show the spatial distribution of contaminant levels while also showing the progression over time. At the same time, we didn't want to overload the viewer with information. We ended up splitting the time series into multiple maps. Limiting to a few counties in the Triangle region helped focus our scope while still giving us enough room to see all the variation.

Outcome

Previous work found a seasonal trend with nitrates, associated with fertilizer use. Our data did not support any seasonal trends. Instead, it appears to be a long-term, consistent source of pollution. Based on the spatial trends, we found that high levels were associated with livestock production, particularly pig farms. We suspect that long-term leakage from faulty waste lagoons could be responsible for much of the nitrate and nitrite pollution. Nutrient pollution like nitrates can cause algal blooms that are a danger to both ecological and human health. Models like ours can be used to target pollution sources and create mitigation strategies.